MANAGING USERS AND PERMISSIONS IN LINUX

USER'S RELATED FILE

2./etc/shadow

This file basically stores all the information about the passwords of a user.

Prompt: - {mayur: yj9T$1Yb.8NjABBj: 20553: 0: 99999: 7: 10: 18759}

1. mayur ----------> username
2. yj9T$1Yb ----> Hash format/encrypted format password
3. 20553 ----------> last time password change from 1 January 1970
4. 0 ----------> minimum password age
5. 99999 ----------> maximum password age
6. 7 ----------> warning period
7. 10 ----------> inactivity days
8. 18759 ----------> expiration date

2.Example of hashed format / encrypted format password:

My password Is **HELLO**

A = C
B = D
C = E
D = F
E = G
F = H
G = I
H = J

H E L L O = J G N N Q ----------> **Hash format password of HELLO is JGNNQ**

Note: -

1. Can we decrypt these password ------> **No**
2. Because these password in a hashed format which is not reversible
3. But we can decrypt the password of encryption because these is reversible

Encryption → reversible

Hashing → NOT reversible

3.Last Time Password Change From 1 Jan 1970

1.echo $(($(date +%s)/86400)) ---------> 20553

NOTE: -

1.This is calculated from 1 January 1970

2. 86400 means 86400 seconds in one day

1 day 24 hours

1 hour 60 minutes

1 minute 60 seconds

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24 * 60 * 60 => 86400 seconds

3. 20553 days passed since 1st January 1970
4. 20553 means = 20553 * 86400 ----------> 1,775,779,200
5. 1,775,779,200 / 86400 ----------> 20553

4.Minimum password age

User can change password immediately

1. chage -l mayur
2. chage -m 2 mayur
3. chage -l mayur
4. su - mayur
5. passwd

{you must wait longer to change your password}

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5.Maximum password age

Password expires after these many days

1. chage -l mayur
2. chage -M 2 mayur
3. chage -l mayur
4. su - mayur

{warning: your password will expire in 2 days}

How to set password to never expire

1. chage -l mayur
2. chage -M -1 mayur

6.Warning period

User gets warning 7 days before expiry

1. chage -l mayur
2. chage -W 7 mayur

7.Inactivity days

After expiry, account is disabled after these days

Example:
1. Password expiry → 30 August
2. Inactivity period → 10 days

30 August
↓
PASSWORD EXPIRES
↓
USER HAVE A CHANCE-TO-CHANGE PASSWORD ON THESE DATES
↓
31 Aug → Day 1
1 Sep → Day 2
2 Sep → Day 3
3 Sep → Day 4
4 Sep → Day 5
5 Sep → Day 6
6 Sep → Day 7
7 Sep → Day 8
8 Sep → Day 9
9 Sep → Day 10
↓
AFTER THAT ACCOUNT BECOMES INACTIVE/DISABLED

1.chage -l mayur
2.chage -I 10 mayur

After the password expires, give the user 10 more days. If the user still doesn't change the password, the account becomes inactive.

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8. Account expiration date

1. The date on which the user account itself expires.
2. After this date, the user cannot normally log in, even if the password is correct.

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1. chage -l mayur
2. chage -E "30 April 2026" mayur

How to Set The Account Expiry to Never

1. chage -l mayur
2. chage -E -1 mayur

GROUP MANAGEMENT

Group management basically a way of creating, modifying, and managing groups of users to control permissions and access to files, directories, and system resources.

Why We Create Groups?
Instead of giving permissions to each user individually, Linux allows you to:

1. Adding multiple users into the group
2. Assigning permissions to the group
3. So, All the users in that group getting the same access

GROUP RELATED FILES

1./etc/group
This file basically stores all the information of the groups

2./etc/gshadow
These file stores all the information of a passwords of a group

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HOW TO CREATE A GROUP

1.groupadd tech
2.gpasswd tech
3.groupdel tech

1./etc/group

This file having all the information about the groups

Prompts: - {tech:x:1008:user1, user2}

1.tech ----------> group name
2.X ----------> Password (it stored in the encrypted form in /etc/gshadow file)
3.1008 ----------> group ID
4.user1, user2 --------> Group Members

1.How to create a group

groupadd tech

2.How to assign password to a group

gpasswd tech

3.How to create a group with GID

groupadd -g 1010 tech

4.How to modify the groups GID

groupmod -g 1020 tech

5.How to add single user into the group

gpasswd -a mayur tech

6.How to add multiple users into the group

gpasswd -M user1,user2 tech

7.How to add users using USERMOD command

usermod -aG tech user1

Note: -

a ---> append

G ---> group

8.How to delete users from the group

gpasswd -d user1 tech

9.How to delete a group

groupdel tech

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10.How to change the group name

groupmod -n <new name> <old name>

groupmod -n devops tech

TYPES OF GROUPS

1.Primary group (Default Group)

1. Every user has one primary group
2. It is assigned at the time of user creation
3. User's files are owned by this group by default
4. Example: usermod -g tech user1

2.Secondary group

1. A user can be part of multiple secondary groups
2. Used to give extra permissions
3. Example: usermod -aG tech user1

2./etc/gshadow

This file having all the information about the group's passwords, Groups Admins and the Group members.

Prompts: - {tech: nfiebvbf: mayur: user1, user2}

1.Tech ----------> group name
2.nfiebvbf --------> group password (encrypted)
3.mayur ----------> Group Admin
4.user1, user2 ---> Group members

1.How to assign admin to a group
gpasswd -A mayur tech

2.How to assign multiple Admins to a group
gpasswd -A user1,user2 tech

2.How to remove admin of a group
gpasswd -A '' tech

3.How to assign password to a group
gpasswd tech

4.How to remove password of a group
gpasswd -r tech

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How to hide a shadow file
pwunconv

How to unhide the shadow file
pwconv

How to hide gshadow file
grpunconv

How to unhide gshadow file
grpconv
